Homework 2 Report: Particle Methods

Ferdinando Giordano

March 24, 2026

1 Introduction

In this homework I simulate a simple agent-based predator-prey system in a two-dimensional periodic square domain. The two types of agents are rabbits and wolves. At each time step, both species move randomly, rabbits may reproduce and die because of age, while wolves may eat nearby rabbits, reproduce after eating, and die from hunger if they do not eat for too long. The main goal is to observe how simple local stochastic rules produce population dynamics over time, and to compare these dynamics with the qualitative behavior expected from the Lotka-Volterra predator-prey model.

2 Implementation Setup

The simulation starts from the following setup:

- Domain: square habitat with size $L = 10$ and periodic boundary conditions.
- Initial populations: $N_r = 900$ rabbits and $N_w = 100$ wolves, randomly distributed in the domain.
- Motion: at each time step, every agent chooses a random direction and moves by a step sampled from a normal distribution with mean 0 and standard deviation σ.
- Rabbit rules: reproduction probability $p_r^r = 0.02$ per time step; death at age t_d^r.
- Wolf rules: a wolf can eat rabbits within cutoff radius $r_c = 0.5$ with probability $p_e^w = 0.02$ for each nearby rabbit; after eating, it reproduces with probability $p_r^w = 0.02$; it dies if it does not eat for $t_d^w = 50$ time steps.
- Simulations are run for a few thousand time steps.

Part (a): Baseline Case

For part (a), I initialized the system with rabbits and wolves randomly distributed in the domain and used $\sigma = 0.5$ for both species. Rabbits were set to die at age $t_d^r = 100$. I then ran the simulation for 3000 time steps and recorded the number of rabbits and wolves over time.

The results show clear predator-prey oscillations. In general, the rabbit population increases first, while the wolf population follows with a delay. When wolves become more numerous, the rabbit population decreases strongly, and after prey becomes scarce, the wolves also decrease because of starvation. This is the main qualitative similarity with the Lotka-Volterra equations, which describe the deterministic continuous limit of predator-prey dynamics:

$$\frac{dR}{dt} = \alpha R - \beta RW, \qquad \frac{dW}{dt} = \delta RW - \gamma W \tag{1}$$

where α is the prey growth rate, β the predation rate, δ the predator reproduction rate from eating, and γ the predator death rate. Both models show cyclic predator-prey behavior with a delayed predator response.

At the same time, the simulation is less regular than the Lotka-Volterra model. The oscillations are not smooth, and in my run the wolves eventually went extinct. This is expected in a finite-population stochastic system: oscillations can grow in amplitude until one species hits zero, unlike the neutrally stable orbits of the deterministic Lotka-Volterra model. The present model is stochastic, spatial, and based on discrete agents, while Lotka-Volterra is a deterministic mean-field model.

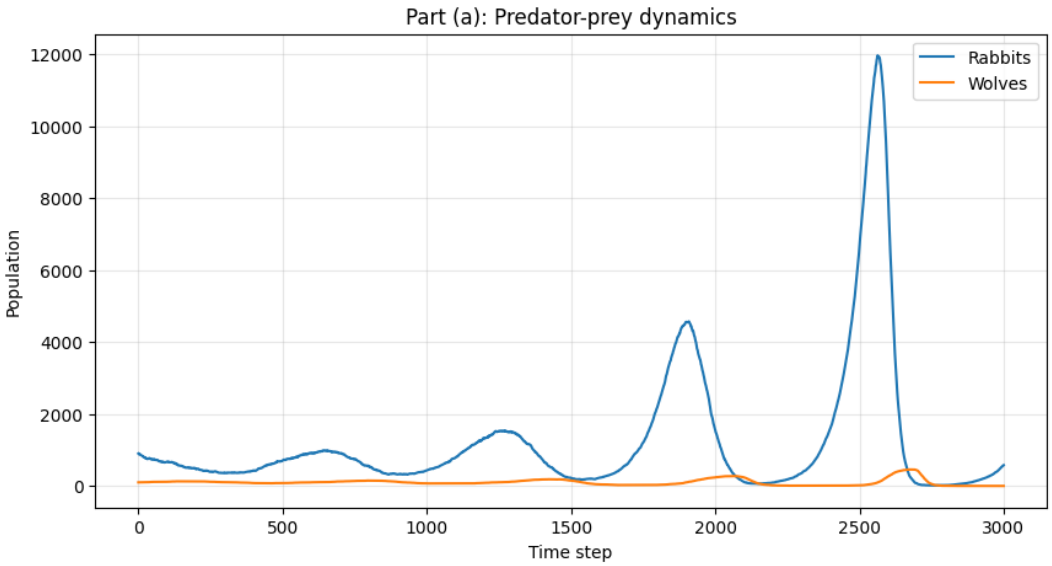

Figure 1: Population dynamics for part (a): rabbits and wolves on the same plot.

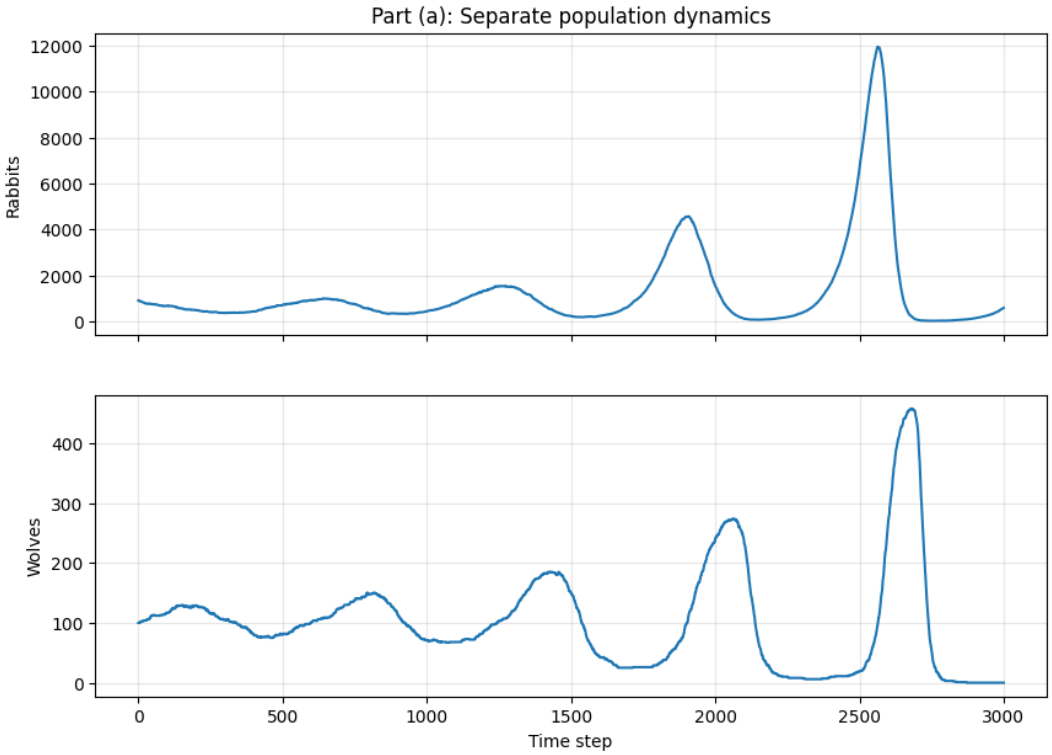

Figure 2: Population dynamics for part (a) shown on separate panels for better readability.

3 Part (b): Shorter Rabbit Lifetime and Different Seeds

In part (b), I repeated the simulations with the same parameters as in part (a), but I reduced the rabbit lifetime to $t_d^r = 50$. I also tested multiple random seeds.

The dependence on the random seed is stronger here because the system is closer to extinction. Small random differences in motion, predation events, and reproduction are enough to produce visibly different trajectories. In all tested seeds the wolves go extinct early, as the reduced rabbit lifetime leaves too little prey to sustain the predator population. What happens after wolf extinction depends on the seed: in seeds 1 and 42 the rabbit population recovers and stabilizes at a low level once predation pressure is removed, while in seed 7 the rabbits also go extinct. This is a purely stochastic outcome - with no wolves and a short lifespan, the rabbit population can still collapse if it drops low enough that reproduction cannot compensate for age-related deaths before the population recovers.

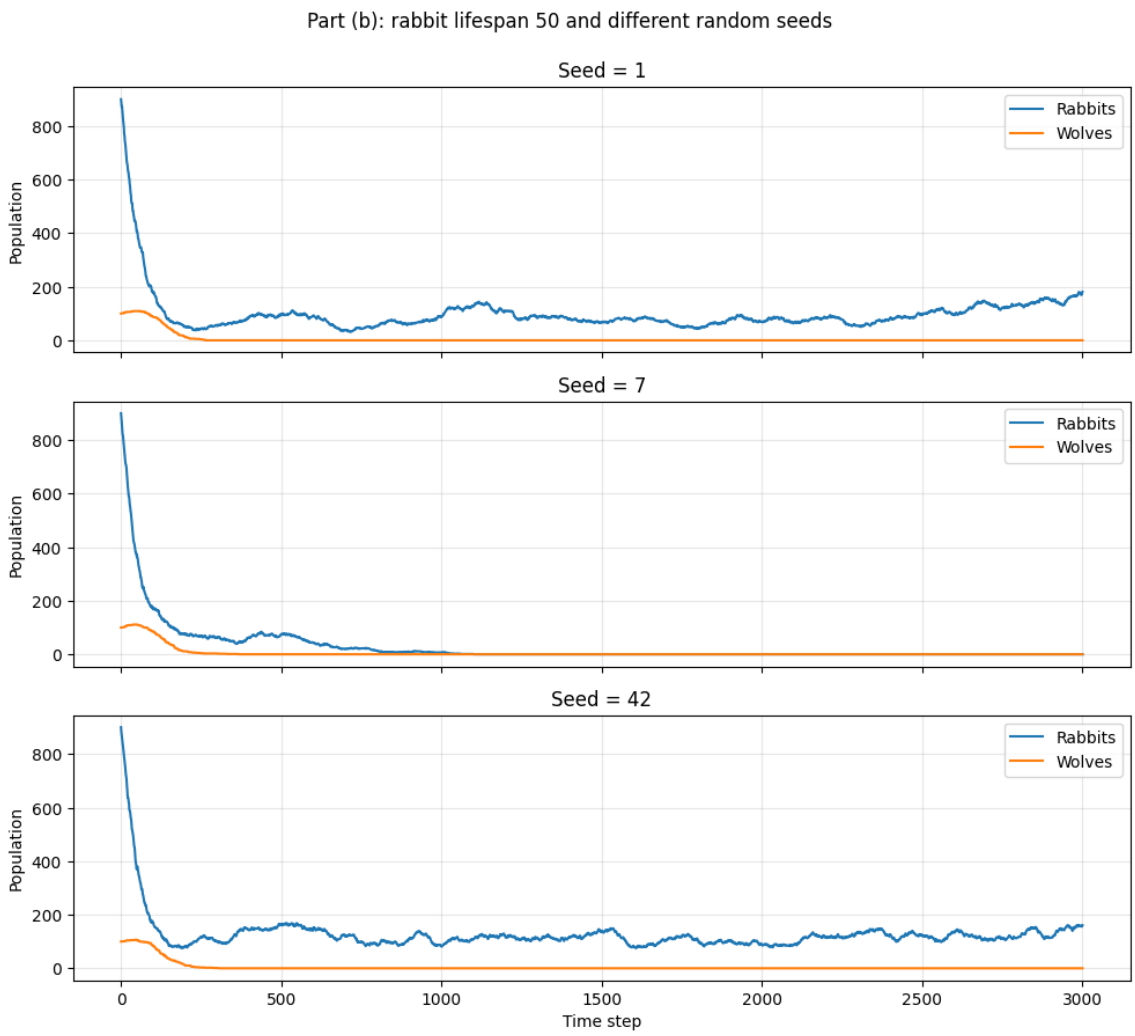

Figure 3: Part (b): full trajectories for different random seeds with rabbit lifetime $t_d^r = 50$.

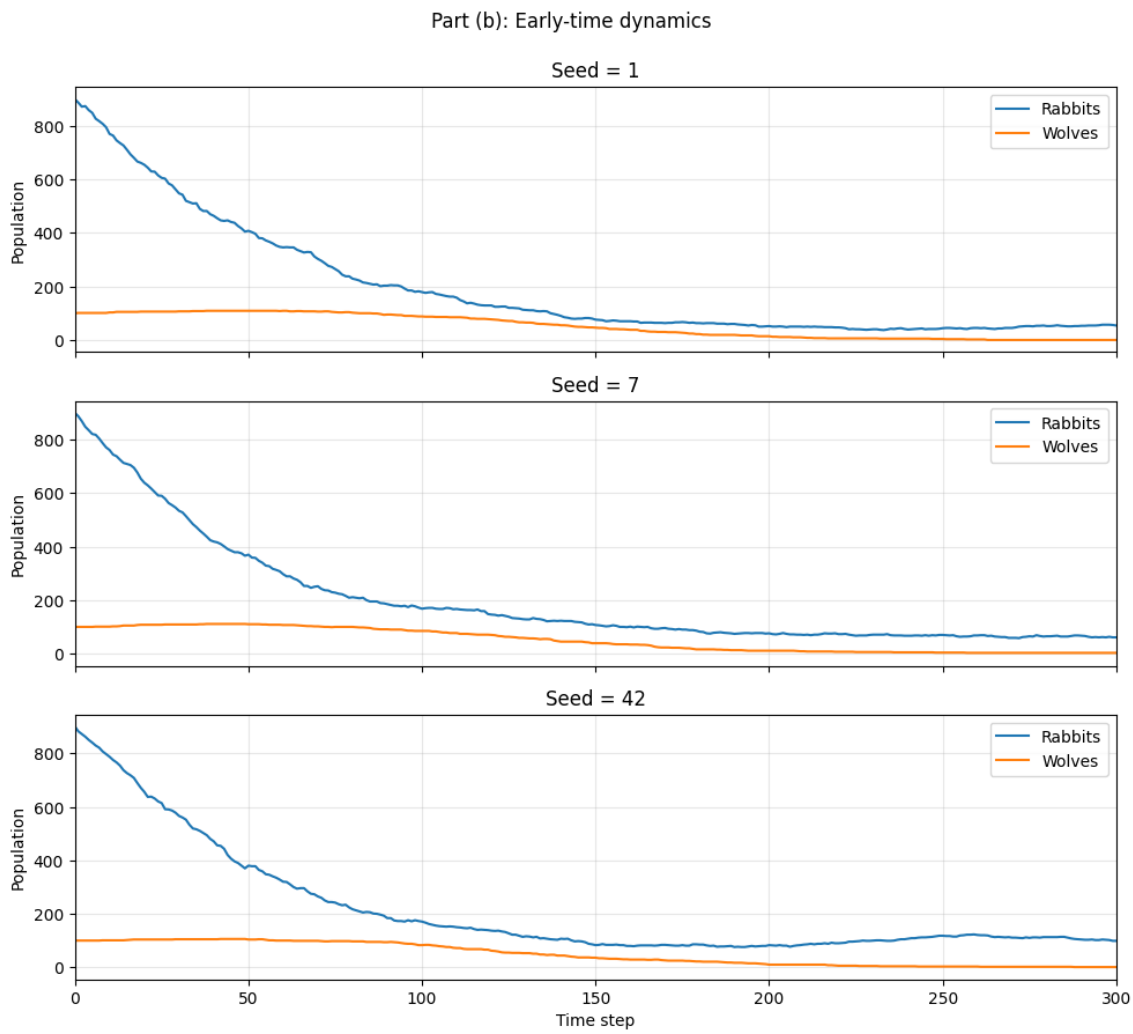

Figure 4: Part (b): early-time dynamics for different random seeds. This makes the rapid wolf decline easier to see.

4 Part (c): Smaller Step Size

In part (c), I kept the same parameters as in part (a), but I reduced the step size parameter to $\sigma = 0.05$ for both rabbits and wolves. I also used $L = 8$ to keep the simulation efficient. The simulation was again run for a few thousand time steps.

The population plot still shows oscillatory predator–prey behavior, but it is now clearer that the interaction is less well mixed than in part (a). Rabbits can grow to very large values before wolves react strongly enough to reduce them, and the predator response appears more delayed and less efficient. This already suggests that reducing the mobility of the agents changes the global dynamics.

The spatial snapshots make the reason more visible. Instead of remaining roughly mixed in the domain, rabbits and wolves form localized groups and non-uniform spatial patterns. At later times, the system is clearly structured in space, with dense rabbit regions and wolves concentrated only in some parts of the habitat. Because interactions depend on local proximity, this spatial clustering becomes important when σ is small.

For this reason, the Lotka–Volterra equations are a weaker approximation in this case. Those equations assume a well-mixed predator–prey system, while here the dynamics are strongly affected by explicit space and local interactions. When the motion is slow, the encounter rate between rabbits and wolves is no longer determined only by the total populations, but also by how agents are distributed in the domain.

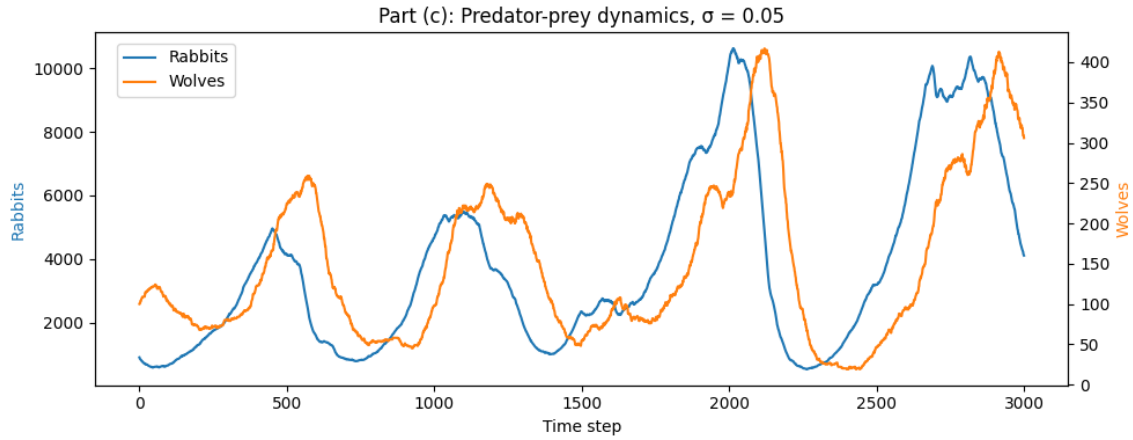

Figure 5: Population dynamics for part (c) with $\sigma = 0.05$. The separate scales make the wolf dynamics easier to see.

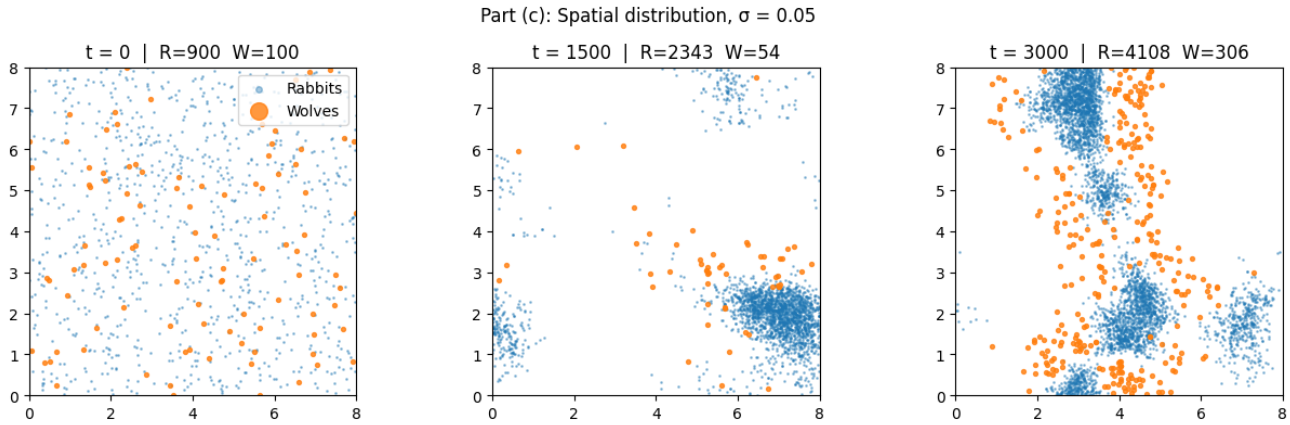

Figure 6: Spatial distribution of rabbits and wolves at selected times in part (c). As time evolves, the system develops visible spatial clustering and becomes far less mixed than in part (a).

5 Conclusion

Overall, the simulations reproduce the main predator-prey feedback expected from simple ecological models: prey grows first, predators follow with a delay, and predator growth later reduces the prey population. This gives a clear qualitative connection with Lotka-Volterra. However, the agent-based model also shows important differences. Because it includes randomness, finite populations, space, and local interactions, the trajectories are more irregular and may lead to extinction. These effects become especially visible when the rabbit lifetime is reduced or when the movement step size is made much smaller.